

X-39MATRIX · DFINITY Developer Grant Application

Target RFP: "Building Post-Quantum Cryptography on ICP"

Submission Date: 2026-06-26 **Applicant:** Jose Luis Olivares Esteban **Email:** grants@x39matrix.org **PGP fingerprint:** C3E062EB251A11851C0B4FFD06870F0655D5BBE8 **Project name:** X-39MATRIX — Sovereign 10-Layer Post-Quantum Protocol on ICP **Repository:** <https://github.com/x39matrix/x39matrix> **Tier requested:** \$25.000 (intermediate) **Existing canisters:** see §3

1. Executive Summary

X-39MATRIX is a production-deployed 10-layer sovereign protocol that runs entirely on Internet Computer canisters, uses threshold-ECDsa on a 13-node ICP subnet to sign Bitcoin mainnet transactions natively, and anchors every public artifact in Bitcoin via OpenTimestamps. The protocol stack already includes 4 mandatory signatures per publication: PGP-Ed25519 + ECDsa-secp256k1 + ML-DSA-87 (FIPS-204) + SLH-DSA-SHAKE-256s (FIPS-205) — making it among the first sovereign infrastructures on ICP with full NIST-finalized post-quantum signature stack in production.

This grant funds the integration of Layer 10 (zk-STARK selective disclosure) into ICP canister architecture, enabling post-quantum zero-knowledge proofs natively on Internet Computer — directly aligned with the May 2026 DFINITY RFP "Building Post-Quantum Cryptography on ICP".

2. Why ICP, why this RFP

DFINITY's roadmap explicitly targets Chain Fusion + post-quantum security. Cloudflare and Google are racing to deploy hybrid PQ signatures by 2027; Internet Computer is uniquely positioned to lead because:

- Threshold-ECDsa on ICP already produces sovereign BTC signatures without bridges, custodians, or seed phrases.
- Canister-based execution allows deterministic PQ verification in WASM, exposed via http_request for browser-side proof checking.
- No equivalent project exists today on ICP combining: PQ signatures + zk-STARK transparent proofs + BTC anchoring + reproducible CI.

This grant brings transparent zk-STARK proofs (no trusted setup, hash-based, quantum-resistant via Grover's bound) to ICP — completing the PQ stack with the missing zero-knowledge layer.

3. Proof of Concept (already running)

Existing production canisters (all module_hash public, verifiable on ic.rocks):

Canister ID	Layer	Lang	Module hash (prefix)
arn4r-lqaaa-aaaao-baxwq-cai	HUB (Sovereign Topos, BTC signer)	Rust	e4ba50b8...
b4dy7-eyaaa-aaaao-baxra-cai	L1 Infrastructure	Motoko	a04f2a13...

Canister ID	Layer	Lang	Module hash (prefix)
b3c6l-jaaaa-aaaao-baxrq-cai	L2 Identity (Merkle + ZK-KYC)	Motoko	a740ea69...
akiau-riaaa-aaaao-baxua-cai	L3 Execution (Ed25519)	Motoko	ad721c01...
anjga-4qaaa-aaaao-baxuq-cai	L4 Consensus (tECSA)	Motoko	d9dbfba7...
s4zl3-eiaaa-aaaao-bay3a-cai	L5 Scalability	Motoko	fd1ddbef...
adlli-haaaa-aaaao-baxvq-cai	L6 Omnichain Bridge	Motoko	8b515717...
awm2f-giaaa-aaaao-baxwa-cai	L7 AI Governance (PTU-47)	Rust	b65cc8b9...
bsbvx-7iaaa-aaaao-baxqa-cai	L8 Notarization (corebackend v2.0.0-realcrypto)	Motoko	4709f6a1...
bvatd-sqaaa-aaaao-baxqq-cai	Frontend (x39matrix.org)	Assets	04e565b3...
nsy7t-jiaaa-aaaau-agwra-cai	Public Dashboard / Evidence Portal	Assets	04e565b3...

Real cryptographic milestones (verifiable now, no overclaim): - 2026-06-02: first sovereign tECSA-signed Bitcoin send (txid `b5a881...ede023`). - 2026-06-07 10:59:51 UTC: first triple-PQ-signed manifest (PGP + ECDSA + ML-DSA-87). - 2026-06-08: quadruple-PQ-signed manifest (added SLH-DSA-SHAKE-256s). - 2026-06-17: sovereign DNS migration anchored across 3 BTC blocks. - 2026-06-24: Layer 10 v1.0 specification published (4 artifacts × PGP × OTS). - 2026-06-26: public corpus anchored in 8 unique BTC mainnet blocks (#955155–#955468).

4. What this grant builds

4.1 Milestone A — `layer10-verifier` canister (Rust, ICP)

Deploy a new canister `layer10-verifier` (~candid spec attached) that: - Accepts a Layer-10 zk-STARK proof + public input via `http_request`. - Verifies the proof deterministically using `winter-verifier` ported to ICP-compatible Rust. - Returns `(verified: bool, btc_anchor: Option<BlockHeight>, pq_signatures: [SignatureBundle])`. - Exposes `module_hash` publicly.

Acceptance: A user in any browser can fetch a `.proof` artifact + send it to the canister via HTTPS and get a deterministic verify response in < 200 ms.

4.2 Milestone B — `layer10-prove` canister (Rust + WASM)

Same as A but for the prover. Deploys the Winterfell prover compiled to ICP-compatible WASM (with appropriate memory budget tuning).

Acceptance: Generate a proof for a 4 KB document in < 5 s on a single ICP node.

4.3 Milestone C — `module_hash` reproducibility + audit kit

Publish: - `Dockerfile` reproducing the exact `module_hash` of both canisters bit-for-bit. - `verify.sh` (separate from the existing top-level verifier) targeting the L10 canisters specifically. - Full benchmark table + signed PGP report + anchored in BTC.

Acceptance: Any auditor can rebuild and verify the `module_hash` published on ICP matches what's in the repo.

5. Honesty disclosure (cypherpunk principle)

The DFINITY Foundation should know:

1. **Layer 10 is currently specification-only.** No Rust implementation exists yet. This grant funds creation, not validation of pre-existing code.
2. **Layer 9 (Custody / Shamir 3-of-5) is in BETA,** not production. Documented as such in `/api/security/layers` endpoint.
3. **The public verifier (`PUBLIC_VERIFY_LAYER10.sh`) does NOT claim "51/51 audit passed"** — that was a previous overclaim in older docs, corrected on 2026-06-26 (audit trail anchored in BTC #955467/#955468 via the `PARCHE_VERIFY_SH.md` artifact).
4. **No human security audit has been performed yet.** I'm targeting NLnet NGIO Security Audit Fund + Cure53 for this — this DFINITY grant is for **implementation**, not audit.

I include this section because protocol credibility depends on never overclaiming. The DFINITY foundation's review process should know exactly what they're funding.

6. Why DFINITY should fund this

1. **Direct alignment with the May 2026 PQ RFP.** ML-DSA-87 + SLH-DSA-256s already in production on ICP, and Layer 10 adds zero-knowledge to the stack.
 2. **Showcases ICP's unique strengths:** threshold-ECDSA + canister-based PQ verification + Chain Fusion to BTC.
 3. **Reproducible from day 1.** Every commit is signed PGP, every artifact anchored in BTC. Auditors do not need access to my keys.
 4. **Solo operator, no token, no VC.** Pure infrastructure work financed by grants. Resistant to capture.
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7. Budget breakdown (\$25.000)

Item	Cost
1 Rust senior × 8 weeks (Milestones A + B)	\$14.000
ICP cycles for testnet + mainnet deployment	\$2.000
Benchmark hardware + reproducibility kit	\$2.000
Documentation + Candid spec + Dockerfile	\$4.000
Public workshop / presentation at ICP community call	\$3.000
Total	\$25.000

8. Timeline

- **Week 1–6:** Milestone A (verifier canister).
 - **Week 7–10:** Milestone B (prover canister).
 - **Week 11–12:** Milestone C (reproducibility + benchmark + public presentation).
 - **Week 13:** Public release + BTC anchor of all artifacts.
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9. Contact and verification

- **Email:** grants@x39matrix.org
 - **PGP fingerprint:** `C3E062EB251A11851C0B4FFD06870F0655D5BBE8`
 - **Repository:** <https://github.com/x39matrix/x39matrix>
 - **Verifier (run yourself in 30s):** `curl -sL https://x39matrix.org/PUBLIC_VERIFY_LAYER10.sh | bash`
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Verifiable infrastructure. No bridges. No custodians. No tokens. No overclaim.

Signed: Jose Luis Olivares Esteban This application SHA-256 to be anchored in BTC mainnet upon submission.